LIFESTYLE IMPACT OF PRODUCTIVITY

A MACHINE LEARNING PERSPECTIVE ON MODERN WORK HABITS

MACHINE LEARNING 1

Group 11

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Abstract

Daily life habits, including sleep patterns, stress levels, breaks, and overall social media usage play a significant role in personal productivity. This study examines how these lifestyle factors influence productivity of a person. By understanding these relationships, the research provides insights for improving daily habits, achieving better life balance, and enhancing both personal well-being and actual productivity.

Contents

Abstract 2
Contents 2
List of figures 3
List of tables 3
Introduction 3
Description of the Question 4
Description of the data set 4
Data Preprocessing 5
1. **Data quality assessment** 5
2. **Imputing missing values** 6
Cluster Analysis 10
K-Prototype Clustering 10
Cluster-Based Models vs Global Model 10
Important Results of Descriptive Analysis 11
Feature Screening Using Mutual Information 11
Correlation and Redundancy Check 11
Mutual Information Analysis of Job Satisfaction 11
Feature Binning and Nonlinear Pattern Exploration 12
Interaction Effect: Job Type × Work Hours 12
Nonlinearity Checks 12
Important Result of the Advanced Analysis 13
Model Evaluation 18
Conclusion 20
Appendix 21

List of figures

Figure 1: Features with missing values and their counts and percentages 6
Figure 2: Distributions before and after for Median Imputation 6
Figure 3: Before and After distributions for Missforest method 7
Figure 4 : Before and after distributions for Bayesian MICE method 8
Figure 5 : Statistical Tests for checking the quality of imputation 8
Figure 6 : Statistical Tests for checking the quality of PMM imputation 9
Figure 7 : Elbow value plot 10
Figure 8 : Modeling for each cluster 10
Figure 9 : DBSCAN clusters with FAMD component as axes 10
Figure 10 : Mutual Information Scores of the features with actual_productivity_score as the target . 11
Figure 11 : Spearman's Correlation with job_satisfaction_score 11
Figure 12 : Mutual Information Scores with job_satisfaction_score as the target 11
Figure 13 : Polynomial model for age and actual_productivity_score as the target 13
Figure 14 : Polynomial model for job_satisfaction_score 13
Figure 15 : SHAP for CatBoost model 18
Figure 16: CatBoost model with job_satisfaction_score only and the actual_productivity_score as the target 18
Figure 17: CatBoost, Linear Regression, Ridge, Lasso, Elastic Net with job_satisfaction_score only as the predictor 19
Figure 18: Linear Regression, Ridge, Lasso, Elastic Net, CatBoost and GAM with job_satisfaction_score only as the predictor. 19

List of tables

Table 1 :Descrption of the dataset 5
Table 2 : Evaluation Matrices for linear regression 13
Table 3 : Evaluation matrices of Lasso .Ridge ,ElasticNet 14
Table 4: Evaluation matrices of KNN 14
Table 5: Evaluation matrices of Random Forest 15
Table 6: Evaluation matrices of Random Forest final model. 15
Table 7: Evaluation matrices of Gradient Boosting 16
Table 8: Evaluation matrices of XGBoost 16
Table 9: Evaluation matrices of LIGHTGBM 17
Table 10: Evaluation matrices of CatBoost 17

Introduction

Personal productivity is influenced by daily habits such as sleep patterns, stress levels, work breaks, and social media usage. These lifestyle factors affect focus, efficiency, and overall well-being, making it essential to understand their impact. This study examines how such habits

influence productivity and provides insights for improving daily routines, achieving better life balance, and enhancing actual performance.

Description of the Question

This study seeks to answer: How do daily life habits and lifestyle factors influence an individual's productivity? Specifically, it investigates the impact of sleep patterns, stress levels, work breaks, burnout frequency, social media usage, job satisfaction on actual productivity. The goal is to identify which habits most strongly affect performance and well-being, providing insights for improving daily routines and achieving better life balance.

Description of the data set

- Dataset obtained from ***Kaggle***
- Contains 30,000 records simulating real world behaviour and lifestyle patterns

Variable	**Description**	**Type**
age	Age of the individual	Integer
daily_social_media_time	Daily social media usage	Continuous
number_of_notification	Number of notifications per day	Integer
work_hours_per_day	Daily work hours	Continuous
perceived_productivity_score	Self reported productivity	Continuous
actual_productivity_score (Target)	Actual productivity score	Continuous
stress_level	Stress level score	Integer
sleep_hours	Sleep hours per day	Continuous
screen_time_before_sleep	Screen time before sleep	Continuous
breaks_during_work	Number of work breaks	Integer
coffee_consumption_per_day	Coffee consumption per day	Integer
days_feeling_burnout_per_month	Burnout days per month	Integer
job_satisfaction_score	Job satisfaction score	Continuous
gender	Gender of individual	Categorical
job_type	Type of job	Categorical

social_platform_preference	Preferred social media platform	Categorical
uses_focus_apps	Uses focus apps	Binary
digital_wellbeign_enable	Digital wellbeing enabled	Binary

Table 1 :Descrption of the dataset

Data Preprocessing

1. Data quality assessment

To perform the initial data quality assessment, a YData-Profiling report was generated. This report provides an overall summary of the dataset, including variable types, missing values, distributions, and potential outliers. The purpose of using this report was to quickly understand the structure and condition of the data before preprocessing. It helped identify data quality issues such as missing values, incorrect data types, and unusual values, allowing appropriate preprocessing decisions to be made and ensuring the dataset was suitable for further analysis.

a. Missing value handling

Records with missing values in the target variable were removed, as these observations cannot be used for modeling. Additionally, values of 0 in the *weekly_offline_hours* variable were identified as invalid since they are not realistic in the given context. Therefore, these values were treated as missing to ensure data accuracy. The dataset was also checked for duplicate observations, and no duplicates were found.

b. Outlier Detection

Outlier detection was performed using both univariate and multivariate approaches. First, univariate analysis was conducted to examine the distribution of individual variables and identify extreme values. Then, DBSCAN clustering was applied to detect multivariate outliers by considering the combined behavior of multiple features. Although some observations were identified as outliers, they did not represent unrealistic or incorrect values. Therefore, these observations were retained in the dataset to preserve the natural variability of the data.
However for daily_social_media_time, values ranging from 12 to 18 hours were identified. Since usage above 12 hours per day was considered unrealistic, these values were capped at 12 hours. This was done to reduce the impact of extreme values while preserving the observations.

2. Imputing missing values

	Feature	Missing Count	Missing Percentage
0	weekly_offline_hours	3005	10.873892
1	daily_social_media_time	2540	9.191243
2	job_satisfaction_score	2499	9.042880
3	sleep_hours	2418	8.749774
4	screen_time_before_sleep	2042	7.389180
5	stress_level	1755	6.350642
6	perceived_productivity_score	1485	5.373620

Figure 1: Features with missing values and their counts and percentages

Several features in the dataset contained missing values, with missing percentages ranging from approximately 5% to 11%, particularly in variables such as *weekly_offline_hours*, *daily_social_media_time*, and *job_satisfaction_score*. Since these variables are important for the analysis, removing the affected observations would result in significant data loss. Therefore, imputation methods were applied to estimate and replace the missing values while preserving the overall structure and information of the dataset.

a. Median Imputation

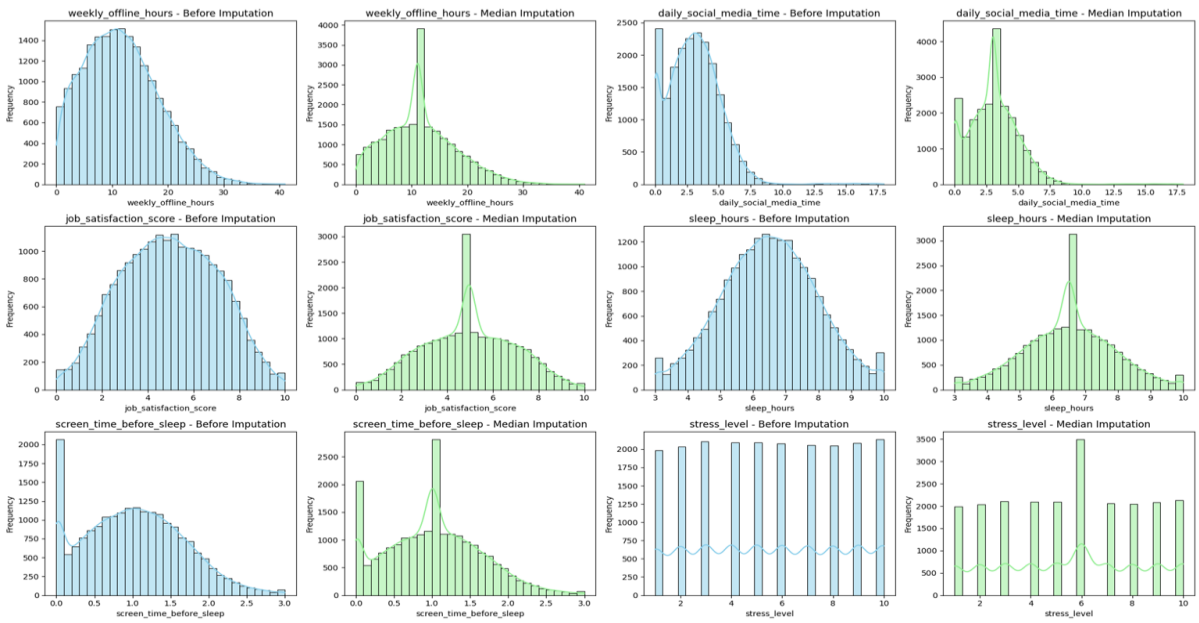


Figure 2: Distributions before and after for Median Imputation

Median imputation replaces missing values with the median of the respective variable. The median is used because it is not affected by extreme values and provides a robust measure of central tendency. This method is simple and easy to implement, making it a common initial approach for handling missing data.

However, median imputation does not consider relationships between variables and may reduce the natural variability of the data. In this study, median imputation resulted in visible spikes in the data distribution at the median value, indicating distortion of the original

distribution. Therefore, this method was considered unsuitable, and more advanced model-based imputation methods were explored next.

b. MICE and MissForest

Model-based imputation methods estimate missing values by learning relationships among variables in the dataset. The variable with missing values is treated as the target, while the remaining variables are used as predictors to build a model. The trained model is then used to predict and replace the missing values, allowing the method to preserve relationships between variables and maintain the overall data structure.

- **MICE (Multiple Imputation by Chained Equations)** imputes missing values by iteratively fitting regression models for each variable with missing data using the remaining variables as predictors.
- **MissForest** uses the Random Forest algorithm to estimate missing values by building multiple decision trees based on the observed data, allowing it to capture complex and nonlinear relationships.

In this study, MICE and MissForest were applied to impute missing values. However, it was observed that the imputed values were highly concentrated around the median, resulting in reduced variability and visible spikes in the distribution.

This occurred because the predictor variables in the dataset showed weak relationships with each other. As a result, the models were unable to learn strong patterns to accurately predict missing values. When predictive relationships are weak (**Shown in EDA**) , model-based methods tend to estimate values close to the overall central tendency of the variable, which is typically near the median.

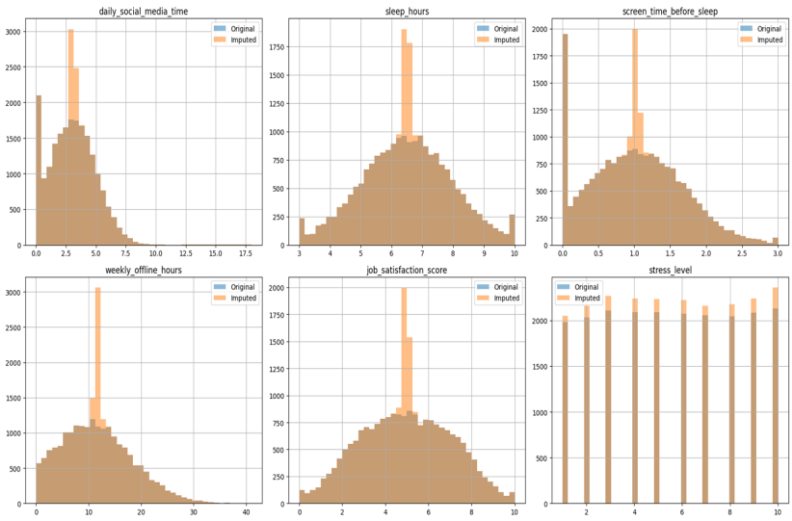


Figure 3: Before and After distributions for Missforest method

Therefore, these methods were not able to preserve the original variability of the data effectively in this case.

The above figure shows the feature distributions before and after imputation using MissForest. We can see clear median spikes. The same kind of results were observed for the MICE imputation method.

c. Bayesian - MICE

This is an extension of MICE that uses Bayesian regression. Instead of producing a single predicted value, it incorporates uncertainty by drawing values from a probability distribution, which can provide more realistic imputations.

Bayesian - Mice imputation method showed better results. Therefore further statistical tests were carried out to check how the distributions differed.

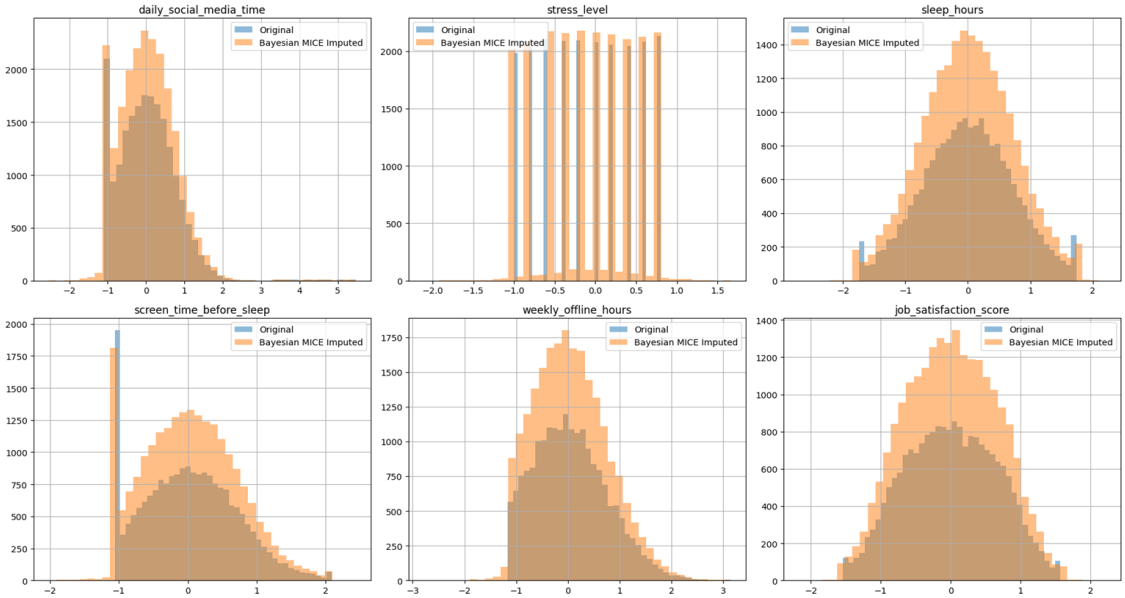


Figure 4 : Before and after distributions for Bayesian MICE method

Bayesian MICE showed improved performance compared to other model-based imputation methods. To further evaluate its suitability, statistical tests were conducted to compare the distributions before and after imputation. The Kolmogorov–Smirnov test showed no significant differences for any variable, indicating that the overall distributions were preserved after imputation. Similarly, the Mann–Whitney U test confirmed that the medians remained unchanged, and the Brown–Forsythe test indicated that the variance was preserved across all features. In addition, Cliff's delta values were negligible for every variable, suggesting that the imputed values had minimal practical difference from the original data. Although the Anderson–Darling test indicated statistically significant differences, this test is highly sensitive to minor changes, especially in large datasets. Overall, these results suggest that Bayesian MICE was able to preserve the key statistical properties of the data and was considered a suitable candidate for missing value imputation.

	Test	Features tested	Significant differences	Interpretation
0	Kolmogorov–Smirnov	6	0	Distribution preserved
1	Anderson–Darling	6	6	Highly sensitive; effect size should be considered
2	Mann–Whitney U	6	0	Median preserved
3	Brown–Forsythe	6	0	Variance preserved
4	Cliff's Delta	6	0	No meaningful practical difference

Figure 5 : Statistical Tests for checking the quality of imputation

d. KNN and PMM

- **K-Nearest Neighbors (KNN) Imputation**

KNN imputation replaces missing values using the values of the most similar observations in the dataset. It identifies the k-nearest neighbors based on feature similarity and imputes the missing value using the average of those neighboring values. This method uses relationships between observations to produce realistic imputations.

- **Predictive Mean Matching (PMM)**

Predictive Mean Matching imputes missing values by first predicting the expected value using a regression model and then selecting an actual observed value from cases with similar predicted values. This ensures that imputed values are realistic and remain within the range of observed data**.**

Both KNN and PMM imputation methods produced similar results. Statistical tests showed no significant changes in the distribution, median, or variance after imputation, indicating that both methods were able to preserve the original data characteristics. The Kolmogorov–Smirnov test confirmed distribution preservation, the Mann–Whitney U test confirmed median preservation, and the Brown–Forsythe test confirmed variance preservation. Additionally, Cliff's delta values were negligible, indicating minimal practical differences between the original and imputed data. Although the Anderson–Darling test detected statistically significant differences, this test is highly sensitive to minor deviations, particularly in large datasets. Given that all other tests confirmed preservation of key statistical properties and the effect sizes were negligible, these differences are not practically meaningful. Therefore, the imputation methods were considered successful in maintaining the integrity of the data.

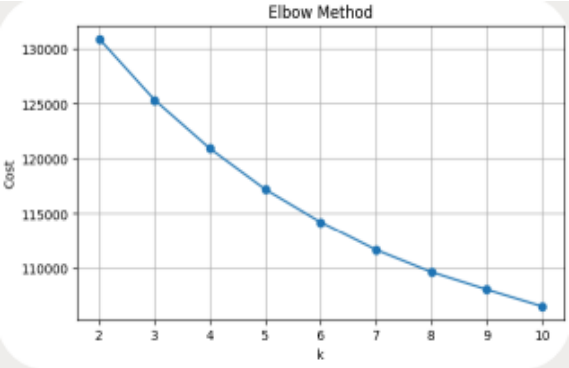
	Test	Features tested	Significant differences	Interpretation
0	Kolmogorov–Smirnov	6	0	Distribution preserved
1	Anderson–Darling	6	6	Highly sensitive; effect size should be considered
2	Mann–Whitney U	6	0	Median preserved
3	Brown–Forsythe	6	0	Variance preserved
4	Cliff's Delta	6	0	No meaningful practical difference

Figure 6 : Statistical Tests for checking the quality of PMM imputation

Although both KNN and PMM preserved the overall distribution and statistical properties of the data, PMM was preferred because it imputes actual observed values rather than averaged estimates. In contrast, KNN imputes missing values using the mean of neighboring observations, which can introduce artificial values that do not exist in the original dataset. By using real observed values, PMM better preserves the natural variability and realism of the data, making it a more statistically appropriate method for imputation.

Cluster Analysis

K-Prototype Clustering



K-Prototype was used because the dataset has both numerical and categorical variables. The elbow curve didn't have a clear, sharp bend, which means there wasn't a strong natural grouping structure. The separation of the clusters looked weak.

Figure 7 : Elbow value plot

```
GLOBAL MODEL PERFORMANCE
============================
R2   : 0.6605
MAE  : 0.8269
RMSE : 1.1024
```

	Cluster	Size	R2	MAE	RMSE
0	0	3993	0.571978	0.874600	1.184134
1	1	3684	0.625190	0.868364	1.131002
2	2	4790	0.653730	0.817242	1.098020
3	3	4841	0.644391	0.830364	1.115817
4	4	4800	0.652738	0.834534	1.107650

Figure 8 : Modeling for each cluster

Cluster-Based Models vs Global Model

To assess practical utility, predictive models were trained both with and without cluster labels. The R^2 improvement was very small (about 0.66 → 0.63). This small difference shows that clustering didn't really improve predictive performance.

DBSCAN Clustering

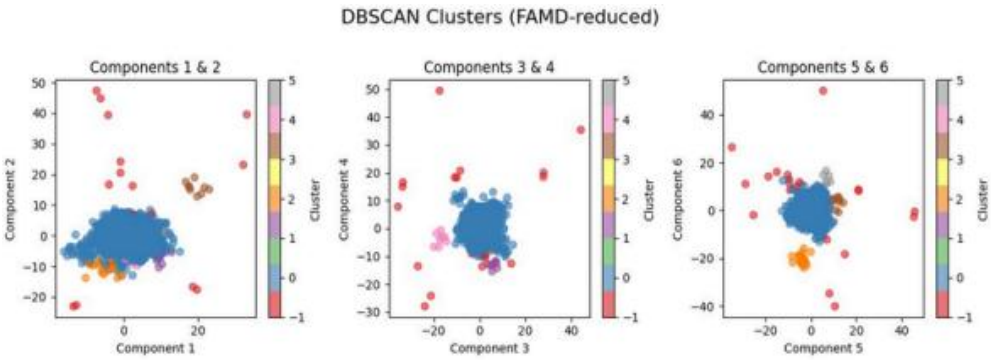


Figure 9 : DBSCAN clusters with FAMD component as axes

DBSCAN was applied to FAMD-reduced components. One dominant large cluster, several smaller clusters, and cluster overlap were high. Therefore, clustering does not provide practical value for this dataset.

Important Results of Descriptive Analysis

Feature Screening Using Mutual Information

```
--- Mutual Information Scores ---
job_satisfaction_score            0.587069
days_feeling_burnout_per_month    0.004876
daily_social_media_time           0.004461
age                               0.004072
screen_time_before_sleep          0.003416
work_hours_per_day                0.002897
weekly_offline_hours              0.002058
job_type                          0.001012
coffee_consumption_per_day        0.000263
sleep_hours                       0.000000
stress_level                      0.000000
number_of_notifications           0.000000
social_platform_preference        0.000000
gender                            0.000000
has_digital_wellbeing_enabled     0.000000
breaks_during_work                0.000000
uses_focus_apps                   0.000000
```

Figure 10 : Mutual Information Scores of the features with actual_productivity_score as the target

Mutual Information (MI) was utilised to assess the relationship between predictors and productivity.

The most important finding was that the job satisfaction score had the strongest link to productivity. On the other hand, most other variables showed very weak dependency.

Correlation and Redundancy Check

Since job satisfaction exhibited significant predictive capability, additional analysis was performed to ascertain its redundancy with other predictors. Correlation analysis showed that job satisfaction and other factors were only weakly related.

This indicates that job satisfaction provides independent predictive information rather than duplicating other variables.

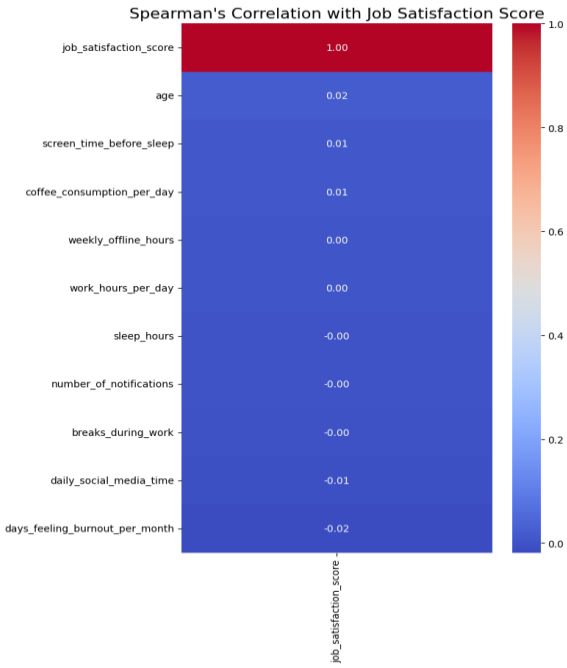


Figure 11 : Spearman's Correlation with job_satisfaction_score

Mutual Information Analysis of Job Satisfaction

```
--- Mutual Information Scores (Target: Job Satisfaction) ---
age                               0.005794
stress_level                      0.004501
work_hours_per_day                0.003028
uses_focus_apps                   0.001834
weekly_offline_hours              0.001781
sleep_hours                       0.000962
number_of_notifications           0.000070
daily_social_media_time           0.000000
social_platform_preference        0.000000
gender                            0.000000
job_type                          0.000000
screen_time_before_sleep          0.000000
breaks_during_work                0.000000
has_digital_wellbeing_enabled     0.000000
coffee_consumption_per_day        0.000000
days_feeling_burnout_per_month    0.000000
```

Figure 12 : Mutual Information Scores with job_satisfaction_score as the target

When job satisfaction was used as the target variable, all other predictors had very weak relationships with it. This indicates that job satisfaction is predominantly autonomous from observable characteristics and may reflect underlying or unobserved behavioural determinants.

Feature Binning and Nonlinear Pattern Exploration

To investigate potential nonlinear relationships, several continuous variables were binned into meaningful categories. These included:

- Screen time before sleep
- Work hours
- Daily social media usage
- Burnout days
- Weekly offline hours
- Coffee consumption

We used statistical tests like ANOVA, Kruskal–Wallis, and Kolmogorov–Smirnov to compare productivity across groups. The results consistently indicated no statistically significant differences. Also, the Mutual Information scores did not get better after binning. This means that there were no nonlinear group-based effects.

Interaction Effect: Job Type × Work Hours

Initial analysis:

Productivity distributions across job types were tested using ANOVA, Kruskal–Wallis, and Kolmogorov–Smirnov. Results showed no significant differences between job types.

However, interaction terms were created: **Job Type × Work Hours.**

Even though each variable had a weak effect, the interaction term between job type and work hours had a predictive effect. This implies that productivity may rely on aggregated behavioral influences rather than singular predictors.

Nonlinearity Checks

Nonlinearity testing was performed to determine whether the relationship between predictors and productivity follows a curved pattern rather than a straight line.

Polynomial regression was used: $\mathbf{Productivity = \beta 0 + \beta 1\ X + \beta 2\ X2 + \epsilon}$

If $\beta 2$ is significant → nonlinear relationship exists.

Nonlinearity Check For Age

Here we fitted two OLS regression models.
Linear: Productivity ~ Age
Polynomial: ~ Age + Age²

From P values we found that the Polynomial term is not significant.

It doesn't improve the R square value so Age is kept as it is.

```
POLYNOMIAL MODEL SUMMARY
                            OLS Regression Results
==============================================================================
Dep. Variable:     actual_productivity_score   R-squared:                       0.000
Model:                                   OLS   Adj. R-squared:                 -0.000
Method:                        Least Squares   F-statistic:                    0.7470
Date:                       Wed, 25 Feb 2026   Prob (F-statistic):              0.474
Time:                               06:25:30   Log-Likelihood:                -45321.
No. Observations:                      22108   AIC:                         9.065e+04
Df Residuals:                          22105   BIC:                         9.067e+04
Df Model:                                  2
Covariance Type:                   nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              4.8170      0.120     40.240      0.000       4.582       5.052
age                0.0067      0.006      1.079      0.281      -0.005       0.019
age_squared   -7.204e-05   7.34e-05     -0.982      0.326      -0.000    7.18e-05
==============================================================================
Omnibus:                     6082.364   Durbin-Watson:                   1.986
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              983.531
Skew:                          -0.011   Prob(JB):                    2.69e-214
Kurtosis:                       1.967   Cond. No.                     2.12e+04
==============================================================================
```

Figure 13 : Polynomial model for age and actual_productivity_score as the target

Nonlinearity Check For Job Satisfaction Score

job_satisfaction_score was the strongest predictor of productivity
Tested whether a nonlinear (polynomial) relationship exists.

```
POLYNOMIAL MODEL SUMMARY (Job Satisfaction)
                            OLS Regression Results
==============================================================================
Dep. Variable:     actual_productivity_score   R-squared:                       0.648
Model:                                   OLS   Adj. R-squared:                  0.648
Method:                        Least Squares   F-statistic:                 2.033e+04
Date:                       Wed, 25 Feb 2026   Prob (F-statistic):               0.00
Time:                               15:40:08   Log-Likelihood:                -33785.
No. Observations:                      22108   AIC:                         6.758e+04
Df Residuals:                          22105   BIC:                         6.760e+04
Df Model:                                  2
Covariance Type:                   nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                      1.3738      0.035     39.082      0.000       1.305       1.443
job_satisfaction_score     0.7274      0.015     48.717      0.000       0.698       0.757
job_satisfaction_squared  -0.0013      0.001     -0.879      0.379      -0.004       0.002
==============================================================================
Omnibus:                     1492.489   Durbin-Watson:                   1.995
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             7379.856
Skew:                          -0.072   Prob(JB):                         0.00
Kurtosis:                       5.827   Cond. No.                         186.
```

Figure 14 : Polynomial model for job_satisfaction_score

Compared two regression models:

- Linear: JobSatisfaction
- Polynomial:
  Productivity ~ Job Satisfaction + (Job Satisfaction)²

The polynomial term is not significant so the relationship is primarily linear.

Important Result of the Advanced Analysis

1. Linear Regression Model

Linear Regression is used as a baseline model to evaluate whether more complex models provide meaningful improvement. If advanced models cannot outperform it, their added complexity is unnecessary.

	Train	Test
MAE	0.8317	0.8384
RMSE	1.1143	1.1314
R2	0.6486	0.6447

Table 2 : Evaluation Matrices for linear regression

Overall, the Linear Regression model explains about 64% of the variation in the data. The MAE and RMSE values are also approximately equal in training and testing. The model shows no evidence of overfitting.

2. Shrinkage Methods

There are 3 shrinkage methods:

- **Ridge Regression**
- **Lasso Regression**
- **ElasticNet**

	Lasso Model		**Ridge Model**		**ElasticNet Model**	
	Train	Test	Train	Test	Train	Test
MAE	0.8330	0.8389	0.8317	0.8384	0.8516	0.8595
RMSE	1.1151	1.1303	1.1143	1.1314	1.1219	1.1372
R2	0.6481	0.6454	0.6486	0.6447	0.6438	0.6410

Table 3 : Evaluation matrices of Lasso .Ridge ,ElasticNet

Ridge, Lasso, and ElasticNet regression models each explain approximately 64% of the variance in the dataset, indicating a moderate and relatively similar level of predictive performance. Among them, the Lasso model demonstrates a slightly higher testing R^2 value, suggesting marginally better generalization to unseen data.

However, since these linear regularization techniques primarily capture linear relationships, we now extend the analysis to tree-based and distance-based models, which are capable of identifying complex non-linear patterns and similarity-based structures in the data, potentially leading to improved prediction accuracy.

3. Distance Based Method

KNN : K-Nearest Neighbors

K-Nearest Neighbors (KNN) is a non-parametric, distance-based supervised learning algorithm that predicts outcomes by identifying the k closest data points in the feature space and using majority voting for classification or averaging for regression, typically measuring similarity using Euclidean distance.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.0000	1.2208	0.0000	1.2208
RMSE	0.0000	1.4698	0.0000	0.4698
R2	1.0000	0.4003	1.0000	0.4003

Table 4: Evaluation matrices of KNN

The baseline model shows perfect training performance but much lower test performance, indicating clear overfitting and limited ability to generalize to unseen data. Although the model was tuned, there is no meaningful improvement in performance, as the test R^2 remains the same and overfitting persists, suggesting that hyperparameter tuning did not enhance the model's generalization ability.

4. Tree Based Methods

Tree-based models are machine learning algorithms that use decision trees to make predictions by splitting data into branches, effectively capturing nonlinear relationships and feature interactions.

a. Random Forest

An ensemble bagging method that builds multiple decision trees in parallel and combines their predictions to reduce variance and overfitting.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.3127	0.8507	0.7593	0.8323
RMSE	0.4172	1.1450	1.0135	1.1239
R2	0.9507	0.6361	0.7092	0.6493

Table 5: Evaluation matrices of Random Forest

Random Forest baseline model shows very high training R^2, but noticeably lower test R^2, indicating overfitting and limited generalization. After hyperparameter tuning with cross validation, training performance decreased while test R^2 slightly improved, showing reduced overfitting and better generalization. A machine learning pipeline was used to incorporate a larger number of trees, improving model stability and ensuring consistent performance across cross-validation folds.

	Tuned Model		**Final Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.7593	0.8323	0.7585	0.8317
RMSE	1.0135	1.1239	1.0129	1.1238
R2	0.7092	0.6493	0.7096	0.6494

Table 6: Evaluation matrices of Random Forest final model.

Random Forest (Final – 800 Trees) increases the number of trees while maintaining similar performance with slightly improved stability and consistency. The tuned and final models demonstrate a better balance between bias and variance compared to the baseline model.

b. Gradient Boosting

A boosting algorithm that builds decision trees sequentially, where each tree corrects the errors of the previous one.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.7925	0.8322	0.8199	0.8302
RMSE	1.0648	1.1235	1.0989	1.1203
R2	0.6791	0.6496	0.6582	0.6516

Table 7: Evaluation matrices of Gradient Boosting

The baseline model shows slightly better performance on training data with lower MAE and RMSE compared to the tuned model, and a higher R^2, indicating it explains about 68% of the variance. On the test set, the performance slightly drops , showing a modest generalization gap. The tuned model has slightly worse training performance but slightly better test performance, suggesting that tuning improved generalization slightly by reducing overfitting, even if the fit to the training data is weaker.

c. XGBoost

An optimized gradient boosting algorithm with built-in regularization for improved efficiency and overfitting control.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.5939	0.8454	0.8139	0.8281
RMSE	0.7905	1.1445	1.0927	1.1211
R2	0.8231	0.6364	0.6621	0.6511
CV R2			0.6530	0.6530

Table 8: Evaluation matrices of XGBoost

Baseline Model shows stronger training performance but a noticeable drop on the test set, along with a larger gap in MAE and RMSE between train and test. This suggests overfitting, where the model fits the training data very well but does not generalize as effectively to unseen data.

In the tuned model, although training performance is lower, the test R^2 improves slightly and is closer to the train R^2, with consistent CV R^2. This indicates better generalization and improved model stability, making the tuned model more reliable for real-world predictions despite slightly weaker training accuracy.

d. LightGBM

A fast, histogram-based gradient boosting method that grows trees leaf-wise for better performance on large datasets.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.5468	0.8443	0.7771	0.8318
RMSE	0.7389	1.1411	1.0423	1.1263
R2	0.8455	0.6386	0.6925	0.6479

Table 9: Evaluation matrices of LIGHTGBM

In the baseline model, it achieves very strong training performance, but test performance drops considerably, with a clear gap between train and test errors. This indicates overfitting, meaning the model captures training data patterns well but does not generalize optimally to unseen data. Tuned Model shows lower training performance but slightly improved test performance and a smaller gap between train and test metrics. This suggests better generalization and improved robustness, making the tuned model more reliable for prediction despite reduced training accuracy.

e. CatBoost

A gradient boosting algorithm specifically designed to handle categorical features efficiently with minimal preprocessing.

	Baseline Model		**Tuned Model**	
	Train	**Test**	**Train**	**Test**
MAE	0.8086	0.8271	0.8086	0.8271
RMSE	1.0860	1.1194	1.0860	1.1194
R2	0.6662	0.6522	0.6662	0.6522

Table 10: Evaluation matrices of CatBoost

Both baseline and tuned models have exactly the same MAE, RMSE, and R^2 values for train and test sets, indicating that hyperparameter tuning did not change the model's behavior or

predictive performance. Therefore, tuning provided no improvement in generalization or accuracy in this case.

Model Evaluation

Based on the evaluation results of all the models, CatBoost achieved the best predictive performance on the test dataset, producing the highest test R2value (0.6522) while maintaining stable MAE and RMSE values. Compared with linear models, distance-based methods, and other tree-based algorithms such as Random Forest, Gradient Boosting, XGBoost, and LightGBM, CatBoost demonstrated slightly better generalization performance. Therefore, **CatBoost was selected as the best performing model for this analysis**.

However, feature importance analysis from the CatBoost model and the results from Exploratory Data Analysis (EDA) both indicate that *job_satisfaction_score* is the only variable with a clear

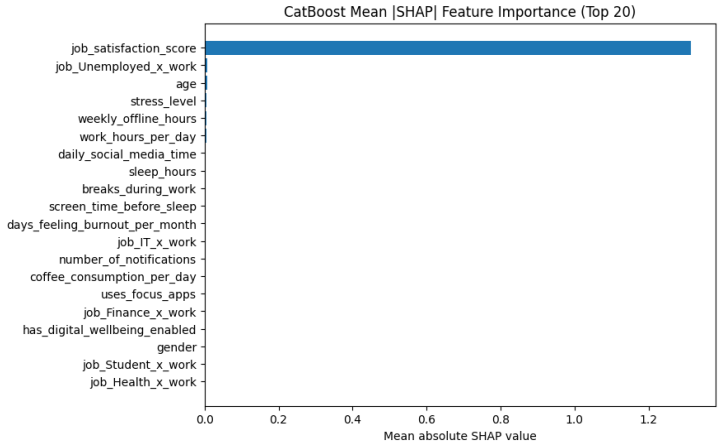


Figure 15 : SHAP for CatBoost model

relationship with *actual_productivity_score*. The remaining features showed minimal contribution to the prediction and did not appear to have a significant impact on the model's performance.

Because of this observation, CatBoost was also trained using only *job_satisfaction_score* as the predictor variable, but the results showed no improvement in performance, confirming that the other variables do not add meaningful predictive value to the model.

Model	Dataset	MAE	RMSE	R²
CatBoost (Job Satisfaction Only)	Train	0.8194	1.1035	0.6553
	Test	0.8254	1.1192	0.6523

Figure 16: CatBoost model with job_satisfaction_score only and the actual_productivity_score as the target

Since job_satisfaction_score is a continuous predictor and actual_productivity_score is also a continuous target variable, linear regression and shrinkage methods such as Ridge, Lasso, and Elastic Net are natural choices for modeling this relationship. These approaches are specifically designed to estimate relationships between continuous variables and provide interpretable coefficients that describe how changes in the predictor influence the response variable.

Model	Dataset	MAE	RMSE	R²
CatBoost (Job Satisfaction Only)	Train	0.8194	1.1035	0.6553
	Test	0.8254	1.1192	0.6523
Linear Regression	Train	0.8321	1.1155	0.6478
	Test	0.8371	1.1299	0.6456
Ridge	Train	0.8321	1.1155	0.6478
	Test	0.8371	1.1299	0.6456
Lasso	Train	0.8326	1.1155	0.6478
	Test	0.8377	1.1299	0.6456
Elastic Net	Train	0.8325	1.1155	0.6478
	Test	0.8376	1.1299	0.6456

Figure 17: CatBoost, Linear Regression, Ridge, Lasso, Elastic Net with job_satisfaction_score only as the predictor.

When these models were trained using only job_satisfaction_score, the results were almost identical across all four models. The training and testing MAE, RMSE, and R^2 values remained very similar, with test R^2 values around 0.6456. This indicates that regularization through Ridge, Lasso, and Elastic Net did not meaningfully change the predictive performance, suggesting that the relationship captured by these linear approaches is essentially the same and that the effect of the predictor is relatively stable.

When CatBoost was trained using only job_satisfaction_score as the predictor, it achieved slightly better predictive performance than the linear models. The model produced lower error values and a higher test R^2 of 0.6523, suggesting that the relationship between job satisfaction and productivity **may include some nonlinear structure** that linear models cannot fully capture.

However, CatBoost is a complex tree-based ensemble algorithm that is generally most advantageous when datasets contain many predictors, interactions between variables, or categorical features that require specialized handling. In this analysis, the predictive relationship is primarily driven by a single continuous variable. In such a setting, the additional complexity of CatBoost provides limited practical benefit and makes the model more difficult to interpret.

To address this, a Generalized Additive Model (GAM) was considered. **GAM extends linear models by allowing smooth nonlinear functions of predictors while maintaining a clear and interpretable structure**. This makes it well suited for situations where the relationship between a continuous predictor and the response may not be strictly linear but still needs to be modeled in an interpretable way.

Model	Dataset	MAE	RMSE	R²
Linear Regression	Train	0.8321	1.1155	0.6478
	Test	0.8371	1.1299	0.6456
Ridge	Train	0.8321	1.1155	0.6478
	Test	0.8371	1.1299	0.6456
Lasso	Train	0.8326	1.1155	0.6478
	Test	0.8377	1.1299	0.6456
Elastic Net	Train	0.8325	1.1155	0.6478
	Test	0.8376	1.1299	0.6456
CatBoost (Job Satisfaction Only)	Train	0.8194	1.1035	0.6553
	Test	0.8254	1.1192	0.6523
GAM (Tuned)	Train	0.8199	1.1050	0.6544
	Test	0.8244	1.1182	0.6529

Figure 18: Linear Regression, Ridge, Lasso, Elastic Net, CatBoost and GAM with job_satisfaction_score only as the predictor.

The tuned GAM produced the best overall performance among the models, achieving the lowest test MAE (0.8241), lowest RMSE (1.1181), and the highest test R^2 (0.6529). Although the improvement compared to CatBoost is relatively small, GAM provides a better balance between predictive accuracy and interpretability. Unlike CatBoost, which behaves as

a black-box ensemble model, GAM explicitly models the smooth nonlinear relationship between

job_satisfaction_score and actual_productivity_score, allowing the effect of the predictor to be visualized and interpreted more clearly.

Therefore, despite CatBoost showing competitive performance, the Generalized Additive Model is preferred for the final analysis because it achieves the best predictive results while providing a more transparent and statistically interpretable representation of the relationship between job satisfaction and productivity.

Conclusion

This study examined how daily lifestyle habits influence individual productivity using a dataset of 30,000 observations representing simulated behavioral patterns. The analysis considered several potential predictors including sleep patterns, stress levels, work breaks, social media usage, burnout frequency, and job satisfaction.

The descriptive analysis revealed that most lifestyle variables showed weak relationships with productivity. Mutual Information analysis, correlation checks, and statistical tests consistently indicated that job_satisfaction_score was the only variable with a meaningful relationship with actual productivity. Other variables showed minimal dependency and did not significantly contribute to predictive performance. Further investigation suggested that job satisfaction was largely independent of the other observed predictors, indicating that it may represent latent behavioral or psychological factors not directly captured by the remaining variables.

Several predictive models were assessed, including linear models, shrinkage methods, distance-based methods, and tree-based ensemble models. Linear Regression, Ridge, Lasso, and Elastic Net yielded nearly equivalent outcomes, accounting for roughly 64% of the variance in productivity. K-Nearest Neighbors exhibited evident overfitting and limited generalization capabilities. Tree-based models, including Random Forest, Gradient Boosting, XGBoost, and LightGBM, displayed moderate performance; however, tuning did not result in significant enhancements.

Among the evaluated models, CatBoost achieved the strongest predictive performance with the highest test R^2 and stable error values. However, feature importance analysis showed that job_satisfaction_score dominated the prediction, while the remaining variables had minimal influence. When CatBoost was retrained using only this variable, model performance remained nearly unchanged, confirming that the other predictors did not add meaningful predictive value.

Because the final predictive relationship was primarily driven by a single continuous variable, the use of a complex ensemble model such as CatBoost provided limited interpretability benefits. Therefore, a Generalized Additive Model (GAM) was applied to capture potential nonlinear effects while maintaining an interpretable modeling structure. The tuned GAM achieved the best overall performance, producing the lowest prediction errors and the highest

test R^2 among the evaluated models. In addition, GAM allows the relationship between job satisfaction and productivity to be visualized through smooth functions, providing clearer insight into the underlying pattern.

However, it is important to highlight that the dataset used in this study is **synthetic** and was generated to simulate realistic lifestyle and behavioral patterns. As a result, the findings should **not be interpreted as real world causal conclusions about productivity or human behavior**. Instead, the analysis should be viewed as a methodological exercise demonstrating how different statistical and machine learning techniques can be applied to explore relationships, handle missing data, and evaluate predictive models.

Future studies using real world observational data may produce different results, particularly if stronger behavioral or organizational relationships exist in actual populations. Therefore, while the analysis and modeling approach presented in this study remain valuable, the conclusions regarding specific predictors of productivity should be interpreted with caution when considering real world applications.

Appendix

Google Colab Link: ML/SL - Final Project.ipynb